

Species-Aware Module Review:

Selenocysteine Biosynthesis & Co-translational Incorporation in

Pseudomonas putida KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Commissioned module:** Bacterial and eukaryotic selenocysteyl-tRNA(Sec) synthesis and UGA→Sec recoding **Resolved local bucket:** KEGG `ppu00450` "Selenocompound metabolism" (module area: other_kegg_pathway) **Evidence base:** Direct target-strain UniProt/genome annotation (UP000000556) + bacterial Sec literature

1. Executive summary

P. putida KT2440 **encodes a complete and demonstrably used bacterial selenocysteine (Sec) pathway**. All four functional steps of the module are satisfied by the **bacterial Sela/SelB route**, and the machinery is validated by a genuine downstream selenoprotein target — formate dehydrogenase-O major subunit **fdoG (PP_0489)**, which carries an annotated **selenocysteine at residue 197**. The eukaryotic route (PSTK, SepSecS, SECISBP2, EEFSEC) is absent, as expected for a bacterium.

The **module is therefore** `covered`/satisfiable, but the **local metadata for this module is substantially misaligned**:

- **Two obligatory module genes are MISSING from the candidate list:** the Sec-specific elongation factor **selB (PP_0494)** and the seryl-tRNA(Sec) ligase **serS (PP_4000)**. tRNA(Sec)/selC (a tRNA gene) is also required and not represented.
- **Ten of the 12 candidate genes are out-of-scope over-propagations:** methionine/cysteine/sulfur-assimilation enzymes (metB, metG, cysD, cysNC, mdeA, metH, metE, PP_4348, PP_4594, PP_4637) pulled in only because the resolved bucket is KEGG **ppu00450** "**Selenocompound metabolism**", a broad map that lumps promiscuous Se-amino-acid analog metabolism with the Sec translational trait.

- KT2440 additionally carries the **distinct selenouridine (SeU) trait** (SelU/YbbB, PP_0822), a tRNA-modification trait that shares the SelD selenophosphate donor but is **not** part of the Sec incorporation module.

Bottom line for curation: mark the four Sec steps covered; add selB, serS, selC; reclassify the methionine/cysteine enzymes as out-of-module; document the boundary problem between the commissioned Sec module and the KEGG "Selenocompound metabolism" bucket.

2. Target-organism pathway definition

In scope (the commissioned module): the co-translational, ribosome-dependent pathway that (1) activates selenium to selenophosphate, (2) charges tRNA(Sec) with serine, (3) converts Ser-tRNA(Sec) to Sec-tRNA(Sec), and (4) recodes an in-frame UGA codon as selenocysteine in a SECIS-dependent context. In KT2440 this is realized entirely by the **bacterial branch**.

Must be kept separate (neighboring processes wrongly co-bucketed here): - **Methionine biosynthesis / one-carbon metabolism** (KEGG ppu00270/ppu00920/ppu04980): metB, metG, metH, metE, PP_4637, mdeA, cystathionine β/γ enzymes. - **Sulfate assimilation** (KEGG ppu00261/ppu00920): cysD, cysNC (ATP-sulfurylase). - **tRNA wobble modification / selenouridine trait:** selU (PP_0822) — a separate Se-utilization trait. These enzymes appear under "Selenocompound metabolism" only because KEGG maps the **free-amino-acid seleno-analogs** (selenomethionine, Se-adenosylselenomethionine, selenocystathionine) through the same sulfur/methionine enzymes. That is **not** the co-translational Sec trait.

Alternate names / database definitions: - KEGG ppu00450 "Selenocompound metabolism" = broad map (analog metabolism + SelD + SelA + SelU), *not* a clean Sec-incorporation module. - The clean concept corresponds to GO **selenocysteine incorporation (GO:0001514)**, **selenocysteine biosynthetic process**, and the "Sec utilization trait (SUT)" of the comparative-genomics literature (Zhang & Gladyshev). - Gene synonyms: SelD = selenophosphate synthetase / SPS / "selenide, water dikinase"; SelC = tRNA(Sec); SelU = YbbB / tRNA 2-selenouridine synthase.

3. Expected step model (bacterial branch) and status in KT2440

#	Module step	Player (bacterial)	KT2440 gene	Status
1	Selenophosphate synthesis	SelD selenophosphate synthetase	seld / PP_0823 (P59392)	covered (in candidate list)
2	Ser-tRNA(Sec) aminoacylation	SerRS (dual Ser/ Sec)	serS / PP_4000 (Q88FT2)	covered — <i>missing from candidate list</i>
—	tRNA(Sec) scaffold	SelC tRNA(Sec)	sclC (tRNA gene)	covered by functional inference — a mature Sec-containing product (fdoG Sec197) cannot form without an active tRNA(Sec); confirm gene locus with tRNAscan-SE/ GtRNAdb
3	Ser→Sec conversion	SelA Sec synthase	sclA / PP_0493 (Q88QJ8)	covered (in candidate list)
4	UGA recoding + Sec-tRNA delivery	SelB EF + SECIS	sclB / PP_0494 (Q88QJ7)	covered — <i>missing from candidate list</i>
Target	Selenoprotein substrate	selenoprotein w/ in-frame UGA + SECIS	fdoG / PP_0489 (Sec197)	present & validated

Eukaryotic branch (PSTK → SepSecS; SECISBP2 → EEFSEC): zero proteome hits → **not_expected_in_target_taxon** for all eukaryotic-route steps.

Genomic context (strong corroboration): the pathway is organized as a co-localized cluster **PP_0489–PP_0494 = fdoG–fdoH–fdoI–fdhE–sclA–sclB**, with **seld** (**PP_0823**) adjacent to **sclU** (**PP_0822**) elsewhere on the chromosome. The physical adjacency of the selenoprotein target (fdoGHI), its maturation factor (fdhE), and the Sec-decoding genes (sclA, sclB) is exactly the operonic arrangement expected of a functional Sec system.

4. Candidate genes and evidence

High-confidence, in-module (keep)

- **sela** — **PP_0493 (Q88QJ8)**. L-seryl-tRNA(Sec) selenium transferase, EC 2.9.1.1; GO:0004125, GO:0001717, GO:0001514. Role: Ser-tRNA(Sec)→Sec-tRNA(Sec). Evidence: sequence/family + operon context (strong). **Covered**.
- **seld** — **PP_0823 (P59392, reviewed Swiss-Prot)**. Selenophosphate synthetase, EC 2.7.9.3; GO:0004756, GO:0016260. Role: selenium activation for BOTH Sec and SeU traits. Evidence: reviewed entry (strong). **Covered**. Caveat: Seld presence alone does not prove Sec (it also feeds SelU); however here Sela/SelB co-presence removes that ambiguity.

In-module but ABSENT from candidate list (must be added)

- **selB** — **PP_0494 (Q88QJ7)**. Selenocysteyl-tRNA-specific elongation factor (EC 3.6.5.-); GTPase; RNA binding; GO:0001514. Obligatory step-4 gene; adjacent to sela. **Promote to full review; add to module**.
- **serS** — **PP_4000 (Q88FT2)**. Seryl-tRNA(Ser/Sec) synthetase, EC 6.1.1.11; GO:0016260. Dual-function housekeeping SerRS that also charges tRNA(Sec). **Add as step-2 gene** (flag: shared with canonical Ser-tRNA charging; not Sec-exclusive).
- **selC / tRNA(Sec)** — required tRNA; not a protein, so absent from the proteome-derived candidate list. **Flag for tRNA-level curation** (tRNAscan / genome check).
- **fdoG** — **PP_0489 (A0A140FVZ1)** and its subunits fdoH/PP_0490, fdoI/PP_0491, maturation factor fdhE/PP_0492. The Sec-containing substrate that justifies the module. **Document as the selenoprotein target**.

Out-of-scope over-propagations (reclassify to sulfur/methionine metabolism)

metB (PP_0659), metG (PP_1097), cysD (PP_1303), cysNC (PP_1304), mdeA (PP_1308), metH (PP_2375), metE (PP_2698), PP_4348 (cystathionine β -lyase), PP_4594 (cystathionine γ -synthase), PP_4637 (MetE-family). None participate in tRNA(Sec) synthesis or UGA recoding. They enter the bucket via KEGG "Selenocompound metabolism" (free seleno-amino-acid analog turnover). Note cysD/cysNC are even primary-bucketed to ppu00261 and metH to ppu04980 in the metadata itself — internal evidence that they are not native to this module.

5. Gaps, ambiguities, and likely over-annotations

- **Metadata gap (high priority):** the candidate list is built from a KEGG *pathway* bucket and therefore captures 2 of ~5 real module genes while missing selB, serS, and selC. This is a **metadata-construction artifact**, not a biological gap.
- **Boundary error (high priority):** `module_needs_revision` at the bucket level — the KEGG ppu00450 "Selenocompound metabolism" mapping is the wrong source for a "Sec biosynthesis + incorporation" module. 10/12 candidates are false-positive module members.
- **SelD ambiguity:** SelD alone is a weak indicator of Sec (it also serves the SeU/SelU trait,

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Here disambiguated by SelA+SelB co-presence.

- **Evidence-quality caveat on the selenoprotein target:** fdoG's Sec197 rests on **imported genome-annotation evidence (ECO:0000313 | EMBL:AMM02774.1, TrEMBL)** — i.e., the genome pipeline translated an in-frame TGA as Sec. Direct experimental confirmation of the Sec residue (or of the in-frame UGA + downstream bacterial SECIS hairpin) is not on record for KT2440.
- **Paralog note:** the cytoplasmic formate dehydrogenase α -subunit **PP_2185** is a **non-Sec (Cys) homolog** — a reminder that "formate dehydrogenase" annotation $\neq$ selenoprotein. Only fdoG/PP_0489 carries Sec.
- **Selenoproteome size:** exactly **one** selenoprotein genome-wide — consistent with an aerobic Gammaproteobacterium (selenoprotein-rich taxa are Deltaproteobacteria/Clostridia;

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6. Module and GO-curation recommendations

Step status calls: - Step 1 (selenophosphate, SelD) → **covered** (PP_0823). - Step 2 (Ser-tRNA(Sec), SerRS) → **covered** (PP_4000) — *add to module metadata*. - Step 3 (SelA route) → **covered** (PP_0493). - Step 4 (SelB insertion system) → **covered** (PP_0494) — *add to module metadata*. - tRNA(Sec)/selC → **covered by functional inference** (the mature fdoG selenoprotein product is obligate proof of an active tRNA(Sec)); flag only for explicit gene-locus annotation

via tRNAscan-SE/GtRNAdb. Direct tRNA-database retrieval was not obtainable in this review. - All **eukaryotic-route steps** (PSTK, SepSecS, SECISBP2, EEFSEC) → **not_expected_in_target_taxon**.

Module-level: - **module_needs_revision** on the bucket-to-module mapping: replace the KEGG ppu00450 source with a curated Sec-incorporation gene set {selD, serS, selA, selB, selC, + selenoprotein targets}. - Recommend a **separate module/annotation for the selenouridine (SeU) trait** (selU/PP_0822 + selD) and keep it distinct from Sec incorporation. - Move metB/metG/cysD/cysNC/mdeA/metH/metE/PP_4348/PP_4594/PP_4637 to methionine/cysteine/sulfur modules.

GO curation: - GO:0001514 (selenocysteine incorporation) and GO:0016260 (selenocysteine biosynthetic process) are appropriately assignable to PP_0493, PP_0494, PP_0823, PP_4000. No new GO terms appear required; the existing selenocysteine-incorporation and selenouridine-modification terms cover the traits.

7. Genes to promote to full **fetch-gene** review

1. **selB** — **PP_0494 (Q88QJ7)** — missing obligatory step-4 gene; highest priority.
2. **serS** — **PP_4000 (Q88FT2)** — missing step-2 gene; flag dual Ser/Sec function.
3. **fdoG** — **PP_0489 (A0A140FVZ1)** — the sole selenoprotein target; verify in-frame UGA + SECIS; note TrEMBL/imported evidence.
4. **selC / tRNA(Sec)** — confirm tRNA gene at genome level (tRNAscan-SE).
5. **selD** — **PP_0823 (P59392)** and **selU** — **PP_0822 (Q88PM7)** — confirm dual-trait assignment and separate SeU trait.

8. Key references

- Fischer et al. 2007, *Biol Chem* — bacterial UGA recoding requires SECIS, SelB, Sec-tRNA^{Sec} formed from Ser-tRNA^{Sec} by SelA using selenophosphate.

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- Serrão et al. 2021 — bacterial Ser→Sec conversion by homodecameric SelA followed by SelB delivery.

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- Silva et al. 2015 — SelA/tRNA(Sec)/SelD/SelB/SECIS ternary-complex biochemistry.

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- Zhang, Turanov, Hatfield, Gladyshev 2008 — SelD required for both Sec and SeU; SelA/SelB define Sec, YbbB defines SeU.

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- Zhang, Romero, Salinas, Gladyshev 2006 — Sec utilization dynamics; selenoprotein-rich taxa are Deltaproteobacteria/Clostridia (aerobes tend to lose selenoproteomes).

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- Peng et al. 2016 — comparative genomics of Se-utilization traits across >5200 bacteria.

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- Shaw et al. 2012 — selA/selB required for FDH selenoprotein biogenesis (link between Sec machinery and FDH).

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- Primary data: UniProt proteome **UP000000556** (*P. putida* KT2440); reviewed entry **P59392** (SelD); **A0A140FVZ1** (fdoG, Sec197, ECO:0000313 | EMBL:AMM02774.1).

Evidence provenance summary

- **Direct target-strain (strong):** presence/annotation of selD, selA, selB, serS, selU, and the fdoG Sec197 residue — all from the KT2440 proteome/genome (UP000000556).
- **Imported/predicted (moderate):** the fdoG Sec assignment (genome-pipeline translation of in-frame TGA; ECO:0000313).
- **Homology/comparative-genomics transfer (contextual):** mechanistic role of each Sec factor and the expectation of a minimal aerobic selenoproteome (from *E. coli*/broad bacterial studies; strong mechanistic transfer because the KT2440 genes are clear orthologs in an intact operon).

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